

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) Which factor most likely caused animals and plants in India to differ greatly from species in nearby southeast Asia?
 - A) India was a separate continent until 45 million years ago.
 - B) The climates of the two regions are similar.
 - C) Life in India was wiped out by ancient volcanic eruptions.
 - D) India is in the process of separating from the rest of Asia.

- 2) Natural selection, operating on which of the following genetic changes, most likely resulted in the loss of ventral spines in modern freshwater sticklebacks?
 - A) silencing of the *Pitx1* gene
 - B) point mutations in the *Pitx1* gene
 - C) deletion of the *Pitx1* gene
 - D) duplication of the *Pitx1* gene

- 3) Which of the following statements best describes the evolution of complex eyes?
 - A) Complex eyes arose fully formed.
 - B) Complex eyes evolved through a series of steps that improved the functioning of the eyes.
 - C) Eyes in molluscs and vertebrates are descended from the same common ancestor.
 - D) The evolution of mollusc eyes was dependent on the evolution of vertebrate eyes.

- 4) Which of the following characteristics were most likely present in the first animals that colonized land?
 - A) ability to resist dehydration and eat producers
 - B) ability to resist dehydration and presence of a spine
 - C) ability to eat producers and support themselves on four limbs
 - D) ability to resist dehydration and support themselves on four limbs

- 5) Which of the following molecules was most likely the first genetic material on Earth?
 - A) oligopeptides located within protocells
 - B) self-replicating RNA molecules with catalytic activity
 - C) DNA produced by reverse transcriptase from abiotically produced RNA
 - D) DNA molecules whose information was transcribed to RNA and later translated in polypeptides

- 6) Many species of snakes lay eggs. However, in the forests of northern Minnesota, where the growing season is short, only live-bearing snake species are present. Which of the following processes most likely resulted in the differences in snake reproduction?
 - A) natural selection leading to exaptations
 - B) species selection
 - C) goal direction in evolution
 - D) sexual selection

- 7) If one organ is an exaptation of another organ, then which of the following statements is most accurate?
 - A) The two organs are homologous.
 - B) The two organs occur in the same species.
 - C) The two organs have the same function.
 - D) The two organs are undergoing convergent evolution.

- 8) Which of the following did all protocells have in common?
 - A) the ability to synthesize enzymes
 - B) RNA genes
 - C) a surrounding membrane or membrane-like structure
 - D) the ability to replicate RNA

- 9) Which of the following statements provides the best evidence that the fossil record is an incomplete record of evolution?
- A) The fossil record is biased for organisms that had hard shells and skeletons.
 - B) The fossil record shows changes in kinds of organisms that lived on earth.
 - C) Fossils document how new organisms come from preexisting organisms.
 - D) The fossil record shows that many animals are extinct.
- 10) The prokaryotic flagellum is a complex structure consisting of several dozen different proteins. Which of the following statements best describes the origin of flagella in prokaryotes?
- A) The need for more complex structure must have driven evolution.
 - B) Ancestral versions of this protein assemblage were either less functional or had different functions than modern prokaryotic flagella.
 - C) Natural selection cannot account for the evolution of the prokaryotic flagellum.
 - D) Flagella could not have arisen by evolution because the structure is too complex to have evolved gradually.
- 11) Four sediment cores were removed from the floor of an inland sea. The sea was in existence intermittently throughout the entire time that terrestrial life has existed. Researchers wish to locate and study the terrestrial organisms fossilized in these cores. In the diagram, the cores are illustrated as vertical columns, with the top of the column representing the most recent strata and the bottom representing the time when land was first colonized by life.

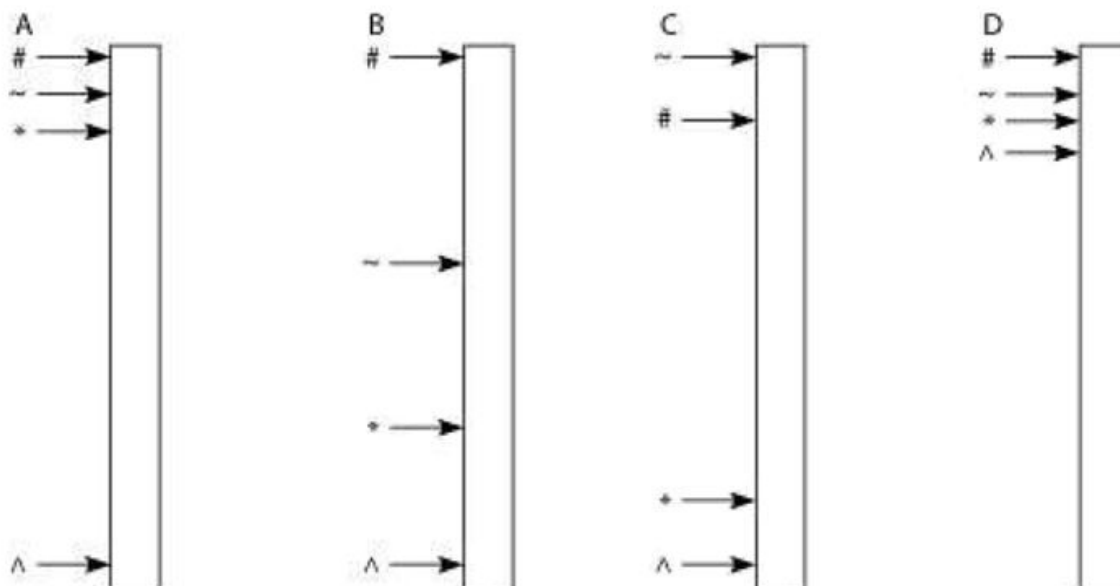
Key:

= carnivorous tetrapods

~ = herbivorous invertebrates

* = plants and fungi

^ = terrestrial cyanobacteria



Which of the following factors would best allow a researcher to assign absolute dates to the fossils in the sediments?

- A) knowledge that the sediments had not been affected by underwater currents during their deposition
- B) equal spacing of fossils throughout the columns
- C) presence of volcanic ash layers between the sedimentary strata
- D) knowledge of the order in which the fossils were deposited

- 12) Which of the following statements is accurate with regard to the Cambrian explosion?
- A) The Cambrian explosion is evidence for the instantaneous creation of life on Earth.
 - B) There are fossils of animals in geological strata that are older than the Cambrian explosion.
 - C) Only the fossils of microorganisms occur in geological strata older than the Cambrian explosion.
 - D) The Cambrian explosion marks the appearance of filter-feeding animals in the fossil record.
- 13) Which of the following methods is the most accurate way to measure the age of a relatively recent fossil?
- A) ratio of carbon-14 to carbon-12
 - B) sequences of fossils in rock strata
 - C) ratio of uranium-238 to thorium-234
 - D) ratio of plutonium-244 to uranium-240
- 14) Which of the following outcomes is the most likely evolutionary effect of duplication of homeotic *Hox* genes?
- A) It allowed the reversal of the head-tail orientation of animals.
 - B) It permitted the evolution of novel forms.
 - C) It reduced morphological diversity of simple forms of life.
 - D) It caused the extinction of major groups.
- 15) Mycorrhizae are symbiotic associations between some fungi and the roots of many plants. Which of the following experiments best would test whether a plant benefits from the presence of the fungus?
- A) Grow many individuals of Plant X in shaded and unshaded conditions. Then compare the average plant size and average number of seeds produced between the two groups.
 - B) Grow many individuals of Plant X with and without its fungus. Then compare the average plant size and average number of seeds produced between the two groups.
 - C) Grow many individuals of Plant X with two different species of fungi. Then compare the average plant size and average number of seeds produced between the two groups.
 - D) Grow many individuals of Plant X and Plant Y without any fungus. Then compare the average plant size and average number of seeds produced between the two groups.
- 16) The temporal fenestra in the skulls in fossils of ancestral mammals changed in shape and size over time. These changes were most likely due to changes in the muscles that closed the jaws. Which of the following research approaches would be most helpful for understanding the evolutionary benefits of these changes?
- A) comparing changes in the fenestra with changes in the jaw hinge
 - B) correlating the fenestra size with changes in the shape of the teeth
 - C) comparing the changes in the fenestra in the fossils leading to mammals with the changes in the fossils leading to modern reptiles
 - D) correlating the strength of jaw closure and the size and hardness of potential prey
- 17) Hawaii is the youngest and most southeastern of the seven largest Hawaiian islands the oldest islands lie to the northwest. Kauai is the oldest and most northwestern big island. Hawaii is also closest to, and Kauai furthest from, the seafloor spreading center from which the Pacific plate originates, which lies about 5,600 km southeast of the island of Hawaii. Assuming equal rates of sedimentation from the islands onto the sea floor, which of the following places is the most likely location of the thickest sediment layer and thus the area with the greatest diversity of fossils above the oceanic crust?
- A) between the island of Hawaii and the seafloor spreading center
 - B) around the base of Kauai
 - C) around the base of the island of Hawaii
 - D) in the area where the islands are most concentrated (highest number of islands per unit surface area)
- 18) Which period had the greatest increase in the number of extant marine animal families?
- A) Permian
 - B) Paleozoic
 - C) Cenozoic
 - D) Mesozoic

- 19) Which of the following statements accurately describes the effect of the Permian and Cretaceous extinctions on the percentage of prey organisms in marine environments?
- A) After the Permian extinction the percentage of prey organisms remained stable, but after the Cretaceous extinction the percentage of prey organisms increased.
 - B) After the Permian extinction the percentage of prey organisms decreased, but after the Cretaceous extinction the percentage of prey organisms continuously increased.
 - C) After the Permian and Cretaceous extinctions, the percentages of prey organisms increased.
 - D) After both the Permian and Cretaceous extinctions, the percentages of prey organisms decreased.
- 20) Which of the following statements best explains the existence of evolutionary trends, such as increasing body sizes among horse species?
- A) A larger volume-to-surface area ratio is adaptive in many mammals.
 - B) In particular environments, similar adaptations can be beneficial to more than one species.
 - C) Evolution generally progresses toward some goal.
 - D) Evolution tends toward increased complexity or increased size.

Answer Key

Testname: UNTITLED1

- 1) A
ID: bio12-- 25.1-28
- 2) A
ID: bio12-- 25.1-38
- 3) B
ID: bio12-- 25.1-50
- 4) A
ID: bio12-- 25.1-23
- 5) B
ID: bio12-- 25.1-3
- 6) B
ID: bio12-- 25.1-49
- 7) A
ID: bio12-- 25.1-44
- 8) C
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- 9) A
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- 10) B
ID: bio12-- 25.1-46
- 11) C
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- 12) B
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- 13) A
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- 14) B
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- 15) B
ID: bio12-- 25.1-27
- 16) D
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- 17) B
ID: bio12-- 25.1-34
- 18) D
ID: bio12-- 25.1-30
- 19) D
ID: bio12-- 25.1-36
- 20) B
ID: bio12-- 25.1-45